



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Civil Engineering

Academic Year 2022-2023 (Even Semester)

Degree, Semester & Branch: IV semester B.E Civil Engineering

Course Code & Title: CE3402 & Strength of Materials

Name of the Faculty member (s): Mr.V.Ragavan

Innovative Practice Description

Unit / Topic: Unit II / Shear Force & Bending Moment Diagram

- Course Outcome: CO2
- Topic Learning Outcome: TLO4
- Activity Chosen: One Minute Paper
- Justification:

Shear force and bending moment diagram is a topic that involves the definition of shear force, bending moment and construction of SFD & BMD for various types of beams. Before starting problem, I want to assess the understanding level of the students that's why I have chosen one minute paper for the above topic.

- Time Allotted for the Activity: 3 minutes

- Details of the Implementation:

i. The one minute paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion.

ii. I asked all students to take a piece of paper and made them to write answers for the questions within one minute.

- CO - PO / PSO mapping:

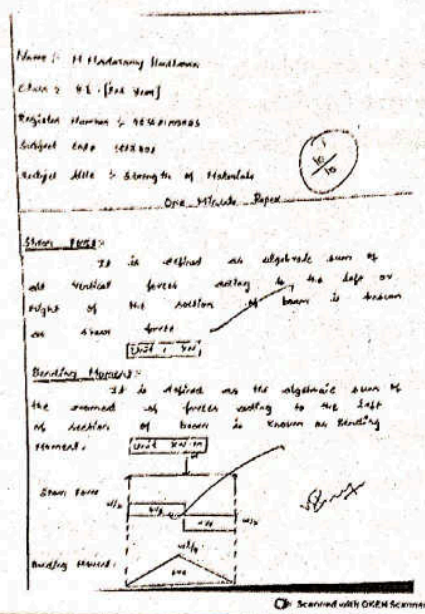
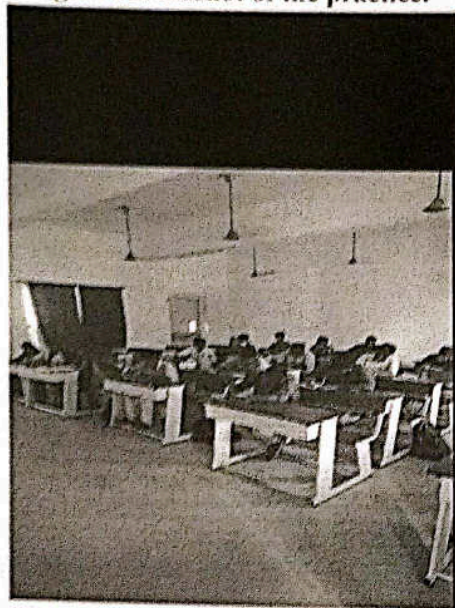
CO	PO1	PO2	PSO2
CO2	3	2	1

(1 - Low 2 - Moderate 3 - High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PSO2
Justification for correlation	Apply the knowledge of basic engineering fundamentals to construct the SFD & BMD for various beams.	Able to determine shear force and bending moment values under various loading conditions and construct SFD & BMD by using fundamental concepts of mechanics and principals of mathematics.	Able to determine shear force and bending moment values under various loading conditions and construct SFD & BMD by Techniques and skills.

• Images / Screenshot of the practice:



• Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

The students felt easy to remember and recollect the concepts involved in the given topic which will make them to attend the questions under this topic in Internal Assessment Test.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- One minute paper activities are an effective way of involving all students in class simultaneously.
- It prompts students to reflect on the day's lesson and provides the instructor with useful feedback.

❖ **Challenges faced in implementation:**

Initially, I have planned the activity for 3 minute but students have taken 5 minutes to write the answers.

References:

- ❖ Rajput.R.K. "Strength of Materials", S.Chand and Co, New Delhi, 2015.
- ❖ Timoshenko.S.B. and Gere.J.M, "Mechanics of Materials", Van Nos Reinhold, New Delhi 1999.


Signature of Faculty Member


HOD